Marshall – Edgeworth Index:

$$P_{01}^{ME} = \frac{\sum p_1 q_0 + \sum p_1 q_1}{\sum p_0 q_0 + \sum p_0 q_1} \times 100$$

$$= \frac{897000 + 970500}{715000 + 772500} \times 100$$

$$= \frac{1867500}{1487500} \times 100$$

$$= 125.55$$

Example 6.10

Calculate a suitable price index form the following data.

Commodity	Quantity	Price	
		2007	2010
X	25	3	4
Y	12	5	7
Z	10	6	5

Solution:

In this problem, the quantities for both current year and base year are same. Hence, we can conlude Kelly's Index price number.

Commodity	q	p_0	p_1	p_0q	p_1q
X	25	3	4	75	100
Y	12	5	7	60	84
Z	10	6	5	60	50
				195	234

Kelly's price Index number:

$$P_{01}^{K} = \frac{\sum p_1 q}{\sum p_0 q} \times 100$$

$$= \frac{234}{195} \times 100$$

$$= 120$$

6.4.2 Weighted average of price relatives

The weighted average of price relatives can be computed by introducing weights into the unweighted price relatives. Here also, we may use either arithmetic mean or the geometric mean for the purpose of averaging weighted price relatives.

The weighted average price relatives using arithmetic mean:

If $p = \frac{p_1}{p_0} \times 100$ is the price relative index and $w = p_0 q_0$ is attached to the commodity, then the weighed price relative index is

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